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ORIGINAL RESEARCH

Kinetic modelling of β-cell metabolism reveals control points in the insulin-regulating pyruvate cycling pathways

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Abstract

Insulin, a key hormone in the regulation of glucose homoeostasis, is secreted by pancreatic β-cells in response to elevated glucose levels. Insulin is released in a biphasic manner in response to glucose metabolism in β-cells. The first phase of insulin secretion is triggered by an increase in the ATP:ADP ratio; the second phase occurs in response to both a rise in ATP:ADP and other key metabolic signals, including a rise in the NADPH:$NADP^+$ ratio. Experimental evidence indicates that pyruvate-cycling pathways play an important role in the elevation of the NADPH:$NADP^+$ ratio in response to glucose. The authors developed a kinetic model for the tricarboxylic acid cycle and pyruvate cycling pathways. The authors successfully validated the model against experimental observations and performed a sensitivity analysis to identify key regulatory interactions in the system. The model predicts that the dicarboxylate carrier and the pyruvate transporter are the most important regulators of pyruvate cycling and NADPH production. In contrast, the analysis showed that variation in the pyruvate carboxylase flux was compensated by a response in the activity of mitochondrial isocitrate dehydrogenase (ICD_m) resulting in minimal effect on overall pyruvate cycling flux. The model predictions suggest starting points for further experimental investigation, as well as potential drug targets for the treatment of type 2 diabetes.

KEYWORDS

beta-cell metabolism, global sensitivity analysis, mathematical model, type 2 diabetes

1 | INTRODUCTION

Pancreatic β-cells integrate a variety of nutrient signals to regulate the secretion of the hormone insulin, which plays an important role in the regulation of nutrient homoeostasis in adipose, liver and skeletal muscle tissue. The failure of β-cell activity, resulting in dysregulated insulin release, in combination with insulin-sensitive tissues becoming insulin resistant, results in the development of type 2 diabetes [1].

The mechanism by which elevated glucose levels trigger the release of insulin within β-cells has not been fully characterised. Glucose-stimulated insulin-secretion (GSIS) is biphasic. The first phase of insulin release is regulated by the K_{ATP}-dependent pathway through which increased glucose metabolism mediates a rise in the ratio of adenosine triphosphate to adenosine diphosphate in the cytosol, causing closure of ATP-dependent potassium (K_{ATP}) channels. The resulting depolarisation of the cell membrane causes an influx of Ca^{2+},

Abbreviations: ACOc, Aconitase (cytosolic); ACOm, Aconitase (mitochondrial); AKD, α-ketoglutarate dehydrogenase; CIC (SLC25 A), citrate carrier; CLc, citrate lyase (cytosolic); CS, citrate synthase; DIC (SLC25 A), dicarboxylate carrier; DPG, 1,3-bisphospho-D-glycerate; F6P, fructose-6-phosphate; FBA, fructose-bisphosphate aldolase; FBP, fructose-1,6-bisphosphate; FM, Fumarase; GAPD, glyceraldehyde 3-phosphate dehydrogenase; GAP, glyceraldehyde 3-phosphate; GK, Glucokinase; GLC, glucose; GT, glucose transporter; IDHc, isocitrate dehydrogenase (cytosolic); IDHm, Iocitrate dehydrogenase (mitochondrial); LDH, lactate dehydrogenase; MDHc, malate dehydrogenase (cytosolic); MDHm, malate dehydrogenase (mitochondrial); MEc, malic enzyme (cytosolic); MEm, malic enzyme (mitochondrial); OGC (SLC25 A), oxoglutarate carrier; PC, pyruvate carboxylase; PDH, pyruvate dehydrogenase complex; PEP, phosphenol pyruvate; PFK, 6-phosphofructokinase; PGP, Bisphosphoglyceratephosphatase; PK, pyruvate kinase; PYC, pyruvate carrier; SCS, succinyl-CoA synthetase; SDH, succinate dehydrogenase.

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triggering the secretion of insulin vesicles [2]. This first-phase of insulin release occurs within the first 10 minutes following glucose stimulation [3]. The second phase of insulin secretion, which is prompted by the K_{ATP}-independent pathway (also known as the amplifying pathway), involves an increase in the ATP:ADP ratio along with other metabolic signals, such as a rise in the NADPH:$NADP^+$ ratio [3, 4].

The K_{ATP} channel-independent pathway relies on the metabolism of pyruvate in the mitochondria. In most cell types, pyruvate feeds the tricarboxylic acid (TCA) cycle via pyruvate dehydrogenase (PDH), which generates acetyl-CoA. β-cells are one of the few cell types that express higher levels of the enzyme pyruvate carboxylase (PC), which provides an alternative route for pyruvate entry to the TCA cycle [5].

There is growing evidence that the compensatory response of PC plays a significant role in the second phase of insulin regulation [5, 6]. The focus of this paper is metabolic pathways involved in the pyruvate cycling process. In particular, we take as primary model outputs the concentration of NADPH and the pyruvate cycling ratio, defined as the ratio of PC flux to the TCA cycle flux [6]. When addressing a complex metabolic network like the TCA cycle, it can be difficult to predict the effects of individual genetic or biochemical perturbations on the entire system. Kinetic modelling provides a framework for addressing networks systematically and quantitatively. Analysis of mathematical models aids in the interpretation of experimental data and can guide the design of further experiments to elucidate the underlying biological process. A number of computational models have been developed to describe aspects of the TCA cycle and GSIS [7–9]. In this paper, we build on these previous efforts to develop a mathematical model of β-cell metabolism that describes pyruvate cycling pathways, with the goal of identifying control points for manipulation of network behaviour, specifically the steady-state pyruvate cycling rate and NADPH concentration.

We describe the development of a detailed model of pyruvate cycling, which we validated against published experimental observations of the pyruvate cycling pathways in insulinoma-derived (INS-1) cell lines and primary rat islets [10, 11]. We successfully fit the model to data from Ronnebaum *et al.* [11] and then validated the model by confirming that it reproduces a wide range of previously published GSIS investigations carried out by ^{13}C isotopomer analysis and siRNA mediated knock-down [6, 12–18]. We performed sensitivity analysis to identify key control points in the pathway. This analysis surveys a broad region in model parameter space to reveal the interaction effects among parameter values. It thus provides a robust identification of the key control points in the pyruvate cycling system. This comprehensive analysis reveals that the pyruvate concentration is heavily influenced by the V_{max} of pyruvate transporter (PYC) and is robust to perturbations in other system features. Considering features of the pyruvate cycling pathways, the V_{max} values of dicarboxylate carrier dicarboxylate carrier (DIC), cytosolic malic enzyme (ME_c), and PC exert significant influence over the pyruvate level, indicating that these are important control points in TCA cycle anaplerosis. Similarly, the NADPH concentration is heavily influenced by these parameters, establishing a further correlation between pyruvate cycling and GSIS.

2 | METHODS

2.1 | Model development

In most cell types, pyruvate feeds the TCA cycle via PDH, which generates acetyl-CoA. However, β-cells express significant quantities of PC, which provides an additional route from pyruvate to the TCA cycle by producing oxaloacetate (OAA_m) from pyruvate (Figure 1). In β-cells, pyruvate flows into mitochondrial pathways through these enzymes in approximately equal proportion [1]. Pyruvate that enters the TCA cycle via PC can readily be recycled back to pyruvate, either directly from OAA_m (via PC) or after further metabolism of OAA_m in the TCA cycle, possibly involving both mitochondrial and cytosolic enzyme activity [18]. The experiments of Lu *et al.* revealed that GSIS is related to PC-catalysed pyruvate cycling, and that PDH-catalysed conversion of pyruvate to acetyl-CoA does not play a significant role in GSIS [6].

PC-based pyruvate cycling involves regeneration of pyruvate from TCA cycle intermediates via three distinct pathways (Figure 1): the pyruvate/malate cycle, the pyruvate/citrate cycle, and the pyruvate/isocitrate cycle. Each cycle begins with the conversion of mitochondrial pyruvate (PYR_m) to oxaloactetate (OAA_m) and ends with the conversion of malate to pyruvate by malic enzyme. (Malic enzyme is active in both the mitochondria and the cytosol. In β-cells, the cytosolic form carries most of the pyruvate cycling flux [19].)

Recent studies have focused on the identification of the metabolic coupling factors that act as signals in the amplifying pathway. These studies provide growing evidence that the pyruvate-cycling pathways generate a metabolic factor that couples increased glucose consumption to insulin release [11, 20–22]. A number of metabolic coupling factors (MCFs) have been proposed, including NADPH, α-ketoglutarate (or its derivatives), and guanosine-5′-triphosphate (GTP) (generated by succinyl-CoA dehydrogenase (SCS)) [1, 23]. Recent observations suggest that NADPH is a key signalling molecule. NADPH is a by-product of all of the pyruvate cycling pathways [24]; it is generated by malic enzyme (a step shared by all three cycles), and by isocitrate dehydrogenase, which is active in the pyruvate/isocitrate pathway. Isocitrate dehydrogenase silencing was found to inhibit GSIS [11].

Several groups have produced computational models describing aspects of the TCA cycle and GSIS. Westermark and colleagues [7] developed a model of mitochondrial nicotinamide adenine dinucleotide (NADH) shuttling (involving 10 metabolites and 19 enzymatic reactions). They validated the model against the findings of Eto *et al.* [25], which characterise the NADH shuttle in β-cells. The TCA cycle has been the subject of many modelling studies. A detailed model of mitochondrial metabolism was developed by Yugi and Tomita [8]. Their model describes 58 enzymatic reactions involving 117

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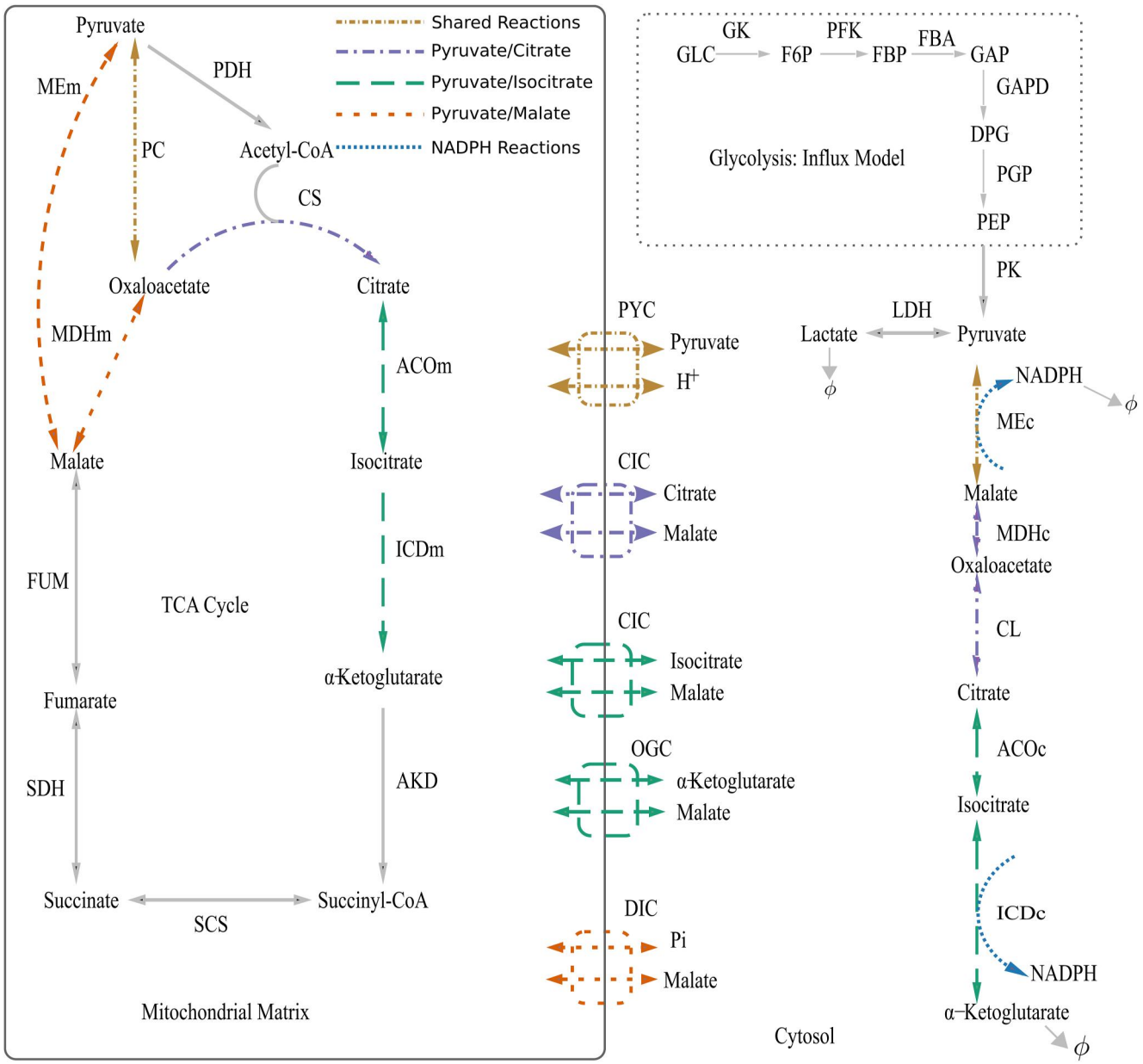

FIGURE 1 Pyruvate cycling pathways in β-cells: Shared reactions (brown): Pyruvate is converted to oxaloacetate by PC. Pyruvate is transported between the cytosol and mitochondria through the pyruvate transporter (PYC). In the cytosol, malate is converted to pyruvate by malic enzyme (MEc). Pyruvate/malate shuttle (orange): Oxaloacetate is converted to malate, while malate is converted to pyruvate through mitochondrial malic enzyme or exported to the cytosol through the dicarboxylate carrier (DIC), where cytosolic malic enzyme converts malate to pyruvate. Pyruvate/citrate shuttle (violet): oxaloacetate is converted to citrate, then transported into the cytosol, where citrate lyase (CL) converts citrate to oxaloacetate, which is then converted to malate by cytosolic malate dehydrogenase (MDH_c). Finally, pyruvate is formed. Pyruvate/isocitrate shuttle (green): Oxaloacetate is converted to citrate and then to isocitrate, both of which are transported into the cytosol. Citrate is converted to isocitrate by ACO_c. Isocitrate can be converted to α-ketoglutarate (α-KG) via ICD_c. Isocitrate is then converted to α-KG by ICD_c. Then α-KG can enter the mitochondria for conversion to malate by tricarboxylic acid (TCA) cycle enzymes, and subsequent conversion to pyruvate by ME_m or ME_c, thus completing the pyruvate cycle. NADPH producing reaction steps are labelled in blue. The kinetic parameters of the six-step influx glycolysis model (boxed) are excluded from the sensitivity analysis below.

metabolites, and incorporates four pathways: the respiratory chain, the TCA cycle, fatty acid β oxidation, and the inner membrane transport system. Jiang *et al.* [9] developed a detailed model of GSIS that describes 44 enzymatic reactions and 59 metabolic state variables. Their model describes five metabolic pathways: glycolysis, the TCA cycle, the respiratory chain, NADH shuttling, and the pyruvate cycle. While these studies involve validation against a range of experimental findings, a systematic corroboration with experimental results on pyruvate cycling, as presented here, has not been attempted. Furthermore, we perform a global sensitivity analysis (GSA) on the pyruvate cycling pathways to identify control points that could not have been identified by local sensitivity or similar linear analysis methods.

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2.2 | Model description

Building on previous modelling efforts, we developed a mathematical model of β-cell metabolism that describes pyruvate cycling. Our model describes the TCA cycle, the pyruvate/malate shuttle, the pyruvate/citrate shuttle, and the pyruvate/isocitrate shuttle, as shown in Figure 1. The model describes 24 metabolites involved in 30 enzymatic reactions. As described in the Supplement (Section 1.4), the bulk of the reaction kinetics are drawn from Yugi and Tomita [8]. These are supplemented with kinetic descriptions from Westermark *et al.* [7] and from the SABIO-RK database [25, 26] (further details below). The model incorporates 129 parameters: 125 of these had nominal values reported in the literature; 4 did not. In preliminary local and global sensitivity analysis focused on pyruvate cycling and NADPH concentration, we found that 34 parameters ranked markedly higher than the others; these included the 4 parameters without reference values. We calibrated these 34 parameters by fitting them to the experimental observations of Ronnebaum *et al.* [11] (details below). Then, as reported below, we tested the model's accuracy by comparing model predictions to qualitative and quantitative observations of system behaviour as reported in the literature on β-cell metabolism [6, 11–18]. While the model exhibits qualitative agreement with most of the available data, we have limited confidence in quantitative accuracy from this calibration. Consequently, we focus our analysis on global sensitivity (complemented by one-at-a-time sensitivity analysis) in our attempt to identify important control points in the pyruvate cycling pathways.

The ordinary differential equation model describes the kinetics of 30 enzyme-catalysed reactions in β-cell metabolism (Figure 1). The model's state variables are the dynamically independent concentrations of 24 metabolite species in two compartments—the mitochondrial matrix and the cytosol (Note, the mitochondrial inter-membrane space is neglected; transport between the mitochondrial and cytosolic compartments is driven directly by concentration differences [20]). Furthermore, all transporters are modelled as bidirectional reactions. One additional species concentration (NADPH) is determined through conservation. Details of the model kinetics and parameter values are described in the supplementary text.

The network input is extra-cellular glucose, which enters the cytosol via the high-capacity, low-affinity glucose transporter-2 (GLUT2). This transport step is modelled as previously reported by Sweet and Matschinsky [28]. Glucose is converted to pyruvate by the six-step glycolysis pathway. The kinetics of all glycolytic reactions were drawn from the SABIO-RK database [27], with some adjustments (details in the supplementary text). The glycolytic model includes the nonlinear relationship between glucose and glycolytic flux, but we excluded the feedback dynamics to balance model complexity. The glycolytic pathway is thus an influx module; its kinetic parameters are not included in the sensitivity analysis presented below. This decision was motivated, in part, by our focus on pyruvate cycling, the regulation of which is understood to be independent of glycolytic activity [3, 4]. The end-product of glycolysis, cytosolic pyruvate, is either converted to lactate by lactate dehydrogenase (LDH) or transported into the mitochondrial matrix. The former process carries much less flux than the latter; LDH activity is weak in β-cells [12]. The TCA cycle operates within the mitochondrial matrix. To keep the focus on pyruvate cycling pathways and managing model complexity we modelled the consumption of NADPH and AKG_c by lumping multiple reactions into a combined outflow rate [7, 28]. All components of the three pyruvate cycling processes—the pyruvate/malate cycle, the pyruvate/citrate cycle, and the pyruvate/isocitrate cycle—are included in the model.

The pyruvate/malate cycle involves the conversion of mitochondrial oxaloactetate OAA_m to malate, via mitochondrial malate dehydrogenase (MDH_m). Mitochondrial malate then follows one of two routes: it can be directly converted to pyruvate by the mitochondrial malic enzyme (ME_m), or it can be transported to the cytosol via the dicarboxylate carrier (DIC) and then converted back to pyruvate by the cytosolic malic enzyme (ME_c).

The pyruvate/citrate cycle begins with OAA_m following its normal route through the TCA cycle: it combines with acetyl-CoA to form citrate via citrate synthase (CS). This mitochondrial citrate can then be converted to isocitrate by mitochondrial aconitase (ACO_m). Mitochondrial citrate and isocitrate are transported into the cytosol by the citrate isocitrate carrier (CIC). The cytosolic form of aconitase (ACO_c) then converts isocitrate to citrate. Citrate lyase (CL) (CL_c) converts citrate to oxaloacetate releasing acetyl-CoA. This cytosolic OAA_c can then be converted to malate by the cytosolic form of malate dehydrogenase (MDH_c). Finally, malate is converted to pyruvate by the malic enzyme, thus completing the cycle. This last step is shared with the pyruvate/malate cycle.

The pyruvate/isocitrate cycle, like the pyruvate/citrate cycle, starts with oxaloacetate being converted to citrate and isocitrate and the subsequent exit of these metabolites from the mitochondria through the CIC. In the cytosol, citrate is converted to isocitrate by ACO_c. Isocitrate is then converted to α-ketoglutarate (α-KG) by cytosolic $NADP^+$-dependent-isocitrate dehydrogenase (ICD_c). α-KG is then transported back into the mitochondria by the oxoglutarate carrier (OGC). Once in the mitochondria, α-KG follows the TCA cycle to be converted to malate, which can then be converted back to pyruvate by the malic enzyme. This last step is shared with the other cycles.

Simulation method: Simulations were carried out in MATLAB® (function ode15s). To ensure the simulations reached a steady-state, we used a trust-region root-finding method to confirm the steady-state conditions (MATLAB® function fsolve). Details of simulations are provided in the supplementary text.

Parameterisation Approach: The reaction kinetics, as well as the bulk of the parameter values, were derived from the previous models of Westermark *et al.* [7] and Yugi and Tomita [8] and the SABIO-RK [26, 27] and Brenda databases [29, 30]. After formulating the model and identifying

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nominal parameter values, we performed a preliminary sensitivity GSA of pyruvate cycling rate and NADPH concentration from which we identified a set of 34 parameters that were consistently ranked highly by all sensitivity ranking methods. We fixed the other 95 parameters at values obtained from the literature (details in the supplementary text), and then fit the values of the 34 high-sensitivity parameters (the resulting values remained near the nominal values obtained from the literature, details in the supplementary text). This calibration was performed by fitting the experimental observations of Ronnebaum *et al.* [11]. That paper provides 32 steady-state metabolite concentration measurements (eight species under four experimental conditions). We will refer to this data as the training set. Parameter values were estimated, under logarithmic transformation, using the simplexSB and simannealSB algorithms provided by the SBTOOLBOX2 software package [31], which minimise the sum of squared errors, weighted by the variability in the data. Details of the fitting procedure, including the bounds used for the parameter search, are reported in the supplementary text. Having found a best-fit to the training set, we then verified the model by comparing its predictions against a range of experimental results on β-cell metabolism, as described in the next section. None of this test set data was used for fitting. The source codes to reproduce model results is available at https://github.com/r2rahul/pyruvatecycle, following the guidelines for publishing reproducible models described by Tiwari *et al.* [32–34]. Further details are provided in the supplementary text.

2.3 | Model analysis methods

The primary aim of this study is to identify network *control points* that have a significant influence over the rate of pyruvate cycling and consequent NADPH production. Our goal is to identify parameters whose values have significant influences over the NADPH concentration and the pyruvate cycling ratio (defined as the ratio of PC flux to the sum of all TCA cycle fluxes, more details provided in the supplementary text section 2), both of which have been implicated as regulators of the second phase of insulin release [3, 4, 6, 11]. Parameters identified as having a significant influence on these model outputs can serve as points of investigation for further experimentation, with the goal of identifying effective drug targets for the treatment of type 2 diabetes.

To identify network control points, we employed one-at-a-time and GSA as described next.

One-at-a-time sensitivity analysis (OAT): We used one-at-a-time parametric sensitivity analysis to investigate the effect of small perturbations in the values of individual parameters. Normalised steady-state sensitivity coefficients were calculated using a first-order forward difference formula (with a 10% deviation from the nominal parameter value). We define S_i as the normalised steady-state sensitivity coefficient with respect to parameter p_i,

$$S_i = \frac{p_i}{F(p_i)} \frac{F(p_i + 0.1p_i) - F(p_i)}{0.1p_i}$$

in which F is the flux or metabolite concentration of interest. We used Richardson extrapolation as described in Elden *et al.* [35] to avoid any inaccuracy due to first-order forward difference approximation.

Global sensitivity analysis: GSA serves two purposes. First, it allows us to explore the nonlinear dependence on model parameters by considering the effects of coincident variations in multiple parameters. Second, it allows a wide exploration of parameter space, which is especially important given the uncertainty of the estimates of the nominal parameter values. Following the recommendation of Zhang and Rundell [36], we used two complementary GSA methods: Partial rank correlation coefficient analysis (with Latin hyper-cube sampling) [37] and the extended Fourier amplitude sensitivity test (eFAST), a variance-based method [38–40]. Both methods are implemented in SBTOOLBOX2 [31, 41].

Variance-based GSA methods address both single-parameter variation—called first-order sensitivity—and multi-parameter variation—called total-effect sensitivity. Comparisons between the first-order and total-effect sensitivities provide insight into nonlinear interactions among parameter-dependent effects. In addition to the sensitivity of key model outputs (fluxes and steady-state concentrations), we also considered the sensitivity of an overall system output, defined as the sum of the squared deviation (from nominal) in all steady-state metabolite concentrations. A complete description of the objective function is provided in the supplementary text.

3 | RESULTS

3.1 | Model training

The model was calibrated against a training set provided by the experiments in Ronnebaum *et al.* [11], in which siRNA specific for the mRNA of ICD_c was used to reduce the activity of ICD_c. To simulate the effect of the siRNA treatment, we reduced the V_{max} parameter of ICD_c by the measured decrease (39.1%) in enzyme activity. Ronnebaum *et al.* [11] collected steady-state measurements of the concentrations of eight metabolites in each of four cases: control and knock-down at low glucose (3 mM) and high glucose (12 mM). Simulating these four cases to steady-state, we fit the model by minimising a sum of squared errors measure of the difference between the steady-state model predictions and the experimentally observed values. Details of the computation are included in the supplementary text. The best-fit model behaviour is shown in Figure 2. The best-fit model parameters are reported in the supplementary text, along with the values of the remaining model parameters, which were taken from the models of Yugi and Tomita [8] and Westermark *et al.* [7], and the Brenda [29, 30] and SABIO-RK [26, 27] databases.

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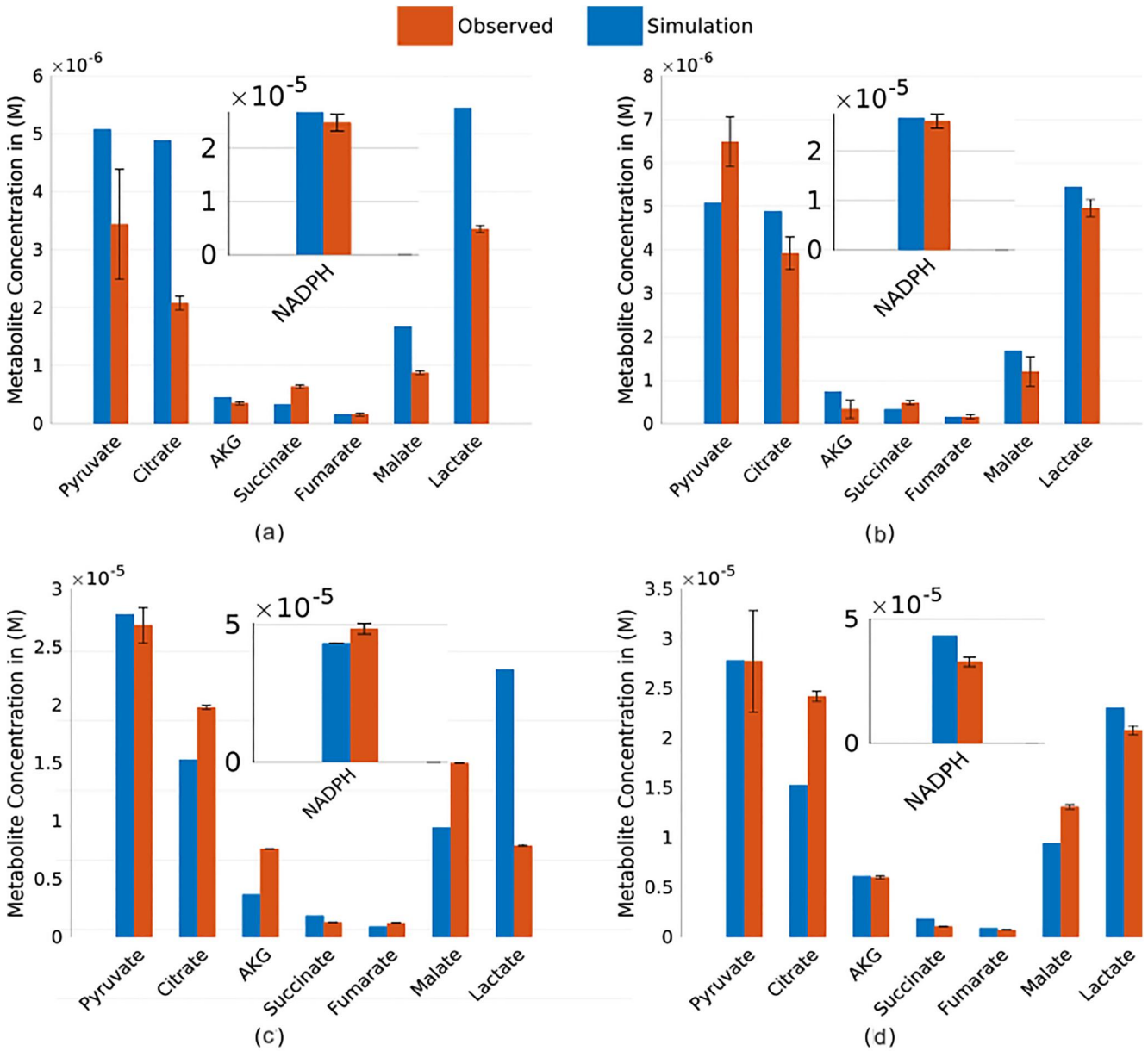

FIGURE 2 Best-fit model steady-state model predictions compared to data from Ronnebaum *et al.* [11]. (a) Low glucose, control. (b) High glucose, control. (c) Low glucose, ICD_c knock-down by 39.1%. (d) High glucose, ICD_c knock-down by 39.1%. NADPH inset is on a different scale.

3.2 | Model testing

We validated the model by comparing model simulations to a series of observations gathered from the literature, as described below. None of these datasets were used in parameter fitting.

Lactate dehydrogenase activity: The model corroborates the experimental observation of Sekine *et al.* [12] that, in comparison with other cell types, β-cells exhibit significantly reduced LDH activity. In the model network, glucose-derived pyruvate is either converted to lactate by LDH or is transported into the mitochondria by the PYC. We compared the flux through these reactions over a wide range of glucose levels (2.5–22 mM) and found that LDH exhibits between 2% and 4% of the PYC flux.

Malic enzyme activity: It has been shown that in β-cells, the cytosolic form of malic enzyme (ME_c) contributes approximately 90% of the total malic enzyme activity in the cell, at glucose concentrations ranging from 3 to 20 mM [13]. Simulations of the model predict that roughly 95% of malic enzyme flux is carried by ME_c, for glucose levels ranging between 2.5 and 22 mM.

Pyruvate carboxylase activity: Radio-isotopic experiments have revealed that in β-cells approximately 40% of glucose-derived pyruvate enters the TCA cycle via PC-catalysed conversion to OAA, with the remainder metabolised to acetyl-CoA via PDH [1]. The model predicts that, within the range of 12–20 mM glucose, approximately 40% of pyruvate enters the TCA cycle through PC. This percentage increases to above 50% at low glucose (3 mM).

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Pyruvate cycling: Using a 13C NMR isotopomer method, Lu *et al.* [6] measured the rate of pyruvate cycling, which they defined as the ratio of PC flux to the overall TCA cycle flux. They found that the anapleurotic flux catalysed by PC is correlated with GSIS, while PDH mediated entry of pyruvate into the TCA cycle is not significantly affected by changes in glucose abundance. We compared the model predictions of the percentage increase in the pyruvate cycling ratio at 3, 6, and 12 mM glucose as observed in the experiments of Lu *et al.* When glucose is increased from 3 to 6 mM, the observed percentage increase in pyruvate cycling ratio was 117% while model simulation shows a 56% increase. When the glucose concentration was raised from 6 to 12 mM, the observed increase in the ratio was 65% while simulations predict an increase of 49%. While the model underestimates the cycling ratio, it correctly captures the increase in cycling ratio with glucose availability. The experiments of Lu *et al.* also revealed that the acetyl-CoA concentration does not increase linearly with glucose availability. When glucose concentration is raised from 3 to 6 mM, the model predicted a 29% increase in acetyl-CoA concentration, whereas a 41% percent increase was observed. Similarly, the model predicted a 16% increase in acetyl-CoA concentration when glucose concentration is raised from 6 to 12 mM, while the observed increase was 70%. Finally, an increase in the glucose concentration from 3 to 12 mM was observed to cause an 86% increase, while the model predicted only a 40% increase. Overall, the model underestimates the percentage increases but correctly captures the acetyl-CoA saturation trend.

Ronnebaum *et al.* [11] studied the effect of an ICD_c knock-down on pyruvate cycling; they measured the pyruvate cycling ratio in wild-type and ICD_c knock-down strains. They observed that the knockdown of ICD_c RNA by 39% caused a reduction of pyruvate cycling to 84 ± 4% of the wild-type ratio at high glucose (12 mM), while at low glucose (3 mM), the knock-down resulted in a slight increase in pyruvate cycling ratio. Our simulations of the knock-down (reducing the V_{max} of ICD_c by 39%) showed no significant effect on the pyruvate cycling ratio at low or high glucose levels.

Citrate lyase knock-down: Joseph *et al.* [15] conducted siRNA-mediated suppression of CL, reducing activity by 75 ± 4%. This resulted in a 52 ± 7% reduction in cytosolic oxaloacetate, and no significant impact on the NADPH:$NADP^+$ ratio (in steady-state, at 16.7 mM glucose). We simulated this experiment by reducing the V_{max} value of CL by 75%. At a steady-state, at 16.7 mM glucose, the resulting change in the NADPH:$NADP^+$: ratio is 0.01%; the cytosolic oxaloacetate concentration drops by 62.8%. The model predicts a more modest drop in OAA_c at lower glucose levels, for example, 54.4% at 3 mM glucose.

Pyruvate carboxylase knock-down: Jensen *et al.* [16] conducted a siRNA-mediated knock-down of PC. In their knock-down strain, PC activity was reduced by 65%, which resulted in a negligible effect on the NADPH:$NADP^+$ ratio. Our model prediction, corresponding to a 65% drop in the V_{max} parameter of PC, shows a 9.9% decrease in the NADPH:$NADP^+$ ratio.

Jensen *et al.* also analyzed the effect of the PC knock-down on the concentrations of TCA cycle intermediates. They found that the concentrations of succinate, malate, and citrate were unaffected by the PC knock-down. Our model simulation of the knock-down predicts succinate, malate, and citrate are reduced by 0.01%, 5.1%, and 0.11% respectively. Furthermore, Jensen *et al.*, found that PC suppression reduced the α-KG concentration by 70% at low glucose (2.5 mM) and 31% at high glucose (12 mM). Our simulation follows a similar trend but showed a strong reduction of 80% and 60% respectively. Additionally, Jensen *et al.* reported increases in the lactate and acetyl-CoA concentrations after PC knock-down. Our simulation captured this trend but showed nominal concentration increases of 9.9% and 11.7% in lactate and acetyl-CoA respectively. This underprediction could be due to the model's description of compensatory activity of malic enzyme, malate dehydrogenase, and PDH. Additional model predictions of the effect of varying PC activity are discussed in the sensitivity analysis section.

Knock-down of cytosolic malic enzyme: To investigate the role of the pyruvate/malate cycle, Ronnebaum *et al.* [17] conducted a siRNA-mediated knock-down of cytosolic and mitochondrial malic enzyme. They found that the knock-down of either enzyme by 75% had no effect on the pyruvate cycling ratio at either 2.5 mM or 12 mM glucose. The corresponding model predictions showed a negligible effect of cytosolic malic enzyme knock-down, but a knock-down of the mitochondrial malic enzyme had a modest effect: a decrease of 0.08% at 2.5 mM glucose but a 9% increase at 12 mM glucose.

Model trends of NADPH concentration profile: In addition to the measurements of NADPH concentration used for model training (Figure 2), Ronnebaum *et al.* [11] also made measurements of the NADPH:$NADP^+$ ratio at a range of glucose concentrations. The model shows reasonable agreement with these findings, as shown in Table 1. However, their observations of the reduction of the NADPH:$NADP^+$ ratio in the ICD_c knock-down strain (as compared to wild-type) were not captured by the model. They observed a decrease in this ratio of 42% at 3 mM and 17% at 12 mM, while the model simulation showed only a negligible decrease.

Sener *et al.* [42] hypothesized that the NADPH:$NADP^+$ ratio can be inferred from the malate:pyruvate ratio or the isocitrate:α-KG ratio. Our model prediction shows that, except for a departure in the ICIT:α-KG ratio at low glucose levels, all these three ratios follow very similar trends, as shown in Figure 3. Finally, Hedeskov *et al.* [13] found that the NADPH:$NADP^+$ ratio increases 125% when glucose is increased from 3 to 20 mM. The model predicts an increase of 71%.

3.3 | Sensitivity results

Sensitivity Rankings: Figures 4 and 5 show the ranked global sensitivities of the model parameters (excluding the kinetic parameters associated with glycolysis) with respect to the two key model outputs (proposed to be directly related to insulin secretion [6, 11]): the steady-state pyruvate cycling ratio and

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TABLE 1 Percentage increase in NADPH concentration in response to an increase in glucose concentration.

Glucose increment (mM)	Experimental (%)	Model Simulation (%)
3–4	33.3	16.4
4–6	35.7	24.6
6–12	50.0	26.3
3–12	125	126

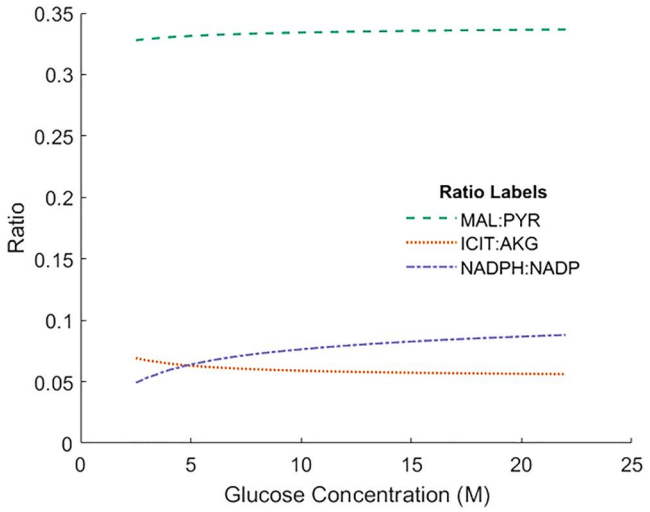

FIGURE 3 Model simulations of the NADPH:NADP$^+$, MAL:PYR, and ICIT:AKG ratios over a range of glucose concentrations.

the steady-state NADPH concentration. Sensitivity rankings of the steady-state pyruvate concentration (cytosolic and mitochondrial), the PC flux, and the flux through cytosolic isocitrate dehydrogenase are included in the supplementary text.

Effect of changes in PC activity: Because it is the first step in all three pyruvate cycling pathways, PC might be expected to exert significant control over the distribution of flux in the network. We calculated sensitivity measures for the effect of perturbations in the V_{max} of PC on steady-state flux for all reactions in the network. Figure 6 shows the five most affected fluxes and the five least affected fluxes.

Further, we compared the sensitivity rankings of eFAST total-effect at low glucose (3 mM) and high glucose (12 mM). We found that parameters of many pyruvate cycling enzymes, such as V_{max} of ME_c, are ranked higher at the high glucose concentration compared to their low glucose rankings. See supplementary text section 4.2 supplementary sensitivity figures. These shifts suggest an increase in the activity of pyruvate cycling enzymes at the higher stimulatory glucose concentration.

4 | DISCUSSION

Given the highly non-linear nature of the model, it is worthwhile to compare the results of different sensitivity analyses to gain insight into model behaviour. By comparing different GSA rankings and One-at-a-time sensitivity coefficients, we can identify key regulators of the metabolic pathway. Comparing the variance-based GSA rankings with the one-at-a-time sensitivity analysis, some differences present themselves, which reveal the extent to which coincident perturbations will have synergistic effects on system behaviour. Likewise, differences between first-order and total-effect sensitivities for other parameters reveal subtleties of interdependence among coordinated perturbations. For example, the V_{max} of ICD_c ranks low in the one-at-a-time analysis, but has a higher ranking in the total-effect GSA. The high total-effect ranking implies that the influence of ICD_c activity level on ICD_c flux is sensitive to the values of other parameters. In the local domain defined by our nominal model parameterisation, the flux is robust to changes in ICD_c activity, but that robustness can be lost when other features of the system are perturbed.

4.1 | Sensitivities of model outputs

Pyruvate cycling ratio sensitivity rankings: Analysing the sensitivity rankings in Figure 4 reveals that the V_{max} of PDH is the most influential parameter in terms of pyruvate cycling. However, a comparison between the first-order and total-effect rankings (panels A and B) reveals that the V_{max}'s of CIC, DIC, and MDH_m are the most significant in terms of modulating the dependence of pyruvate cycling on other parameter values (i.e. their first-order and total-effects sensitivity coefficients are markedly different). These three enzymes are known to be important participants in pyruvate cycling pathways [43].

NADPH sensitivity rankings: Referring to Figure 5, the total effect and one-at-a-time sensitivity rankings (panels A and D) indicate that the NADPH consumption rate has the most influence over the steady-state NADPH concentration; this is consistent with the model structure. Next, comparing the first-order and total-effect rankings (panels A and B) we can draw two conclusions. First, because they are ranked high by both measures, the V_{max}'s of PYC and DIC have a significant impact on the NADPH concentration, regardless of perturbation to the other parameters. Second, the rankings of the V_{max}'s of PDH, PC, and ME_c differ between first-order and total-effect measures, meaning that these parameters are influential in modulating the effect of pathway perturbations on NADPH concentration.

Cytosolic and mitochondrial pyruvate sensitivity rankings: The sensitivity ranking for the steady-state cytosolic pyruvate concentration is shown in the supplementary text. They reveal that the V_{max}'s of the PYC and CS have the most significant impact on pyruvate concentration. Moreover, the agreement between first-order and total-effect rankings implies that PYC and CS dominate the pyruvate concentration, regardless of perturbations to other parts of the pathway.

Pyruvate carboxylase flux sensitivity rankings: The sensitivity rankings of steady-state PC flux, shown in the supplementary text, mirror the results in Figure 4. We find that V_{max}'s of CIC, DIC, and MDH_m are the most significant in terms of modulating the dependence of PC flux on other parameter

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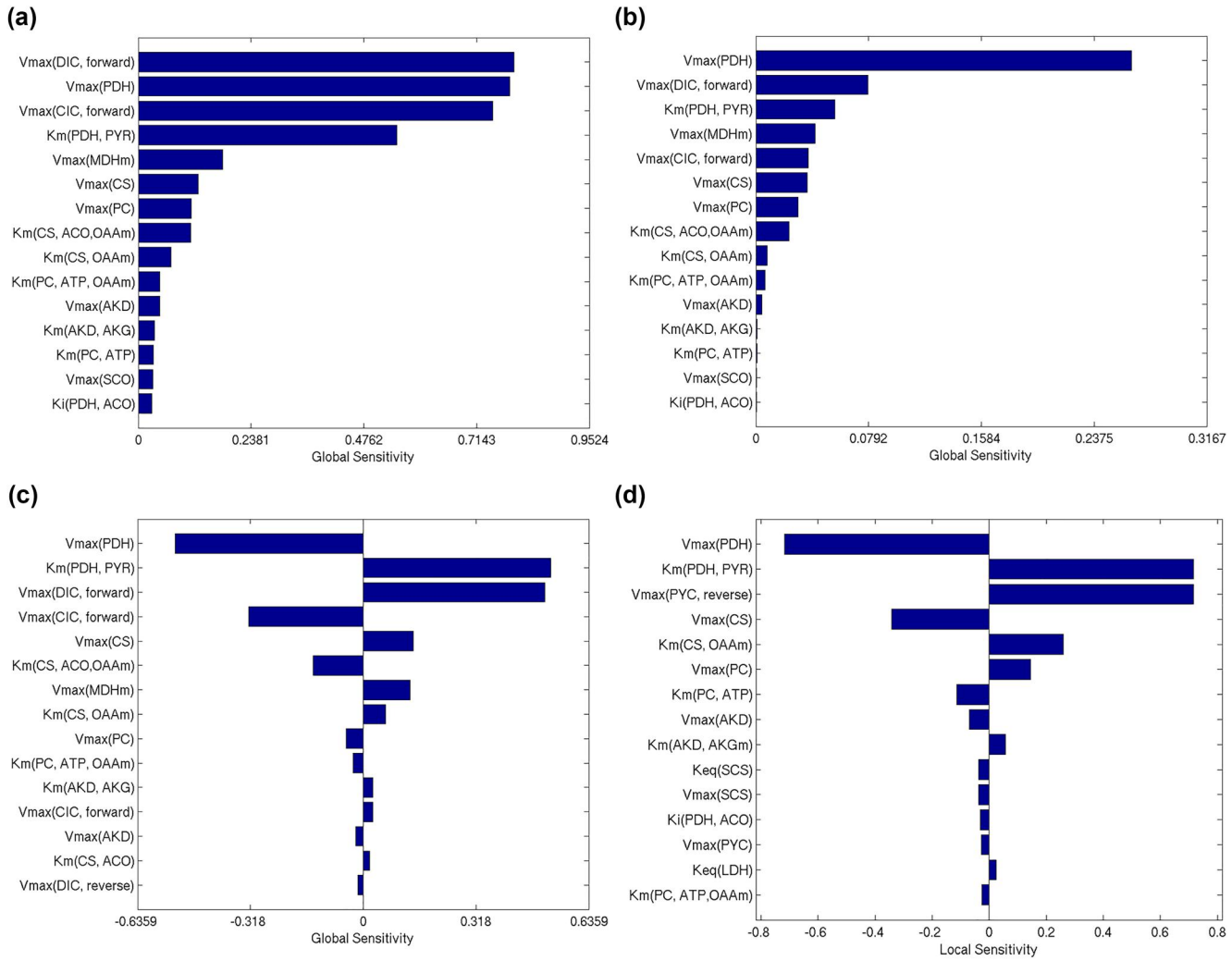

FIGURE 4 Parametric sensitivity ranking of the pyruvate cycling ratio. (a) Extended Fourier amplitude sensitivity test (eFAST) total effect. (b) eFAST first order. (c) Partial rank correlation coefficient (PRCC). (d) One-at-a-time Sensitivity.

values (i.e., their first-order and total-effects are considerably different). However, the V_{max} of PDH has a significant impact on the PC flux, which is consistent with the fact that these two reactions form a branch point.

Cytosolic isocitrate dehydrogenase sensitivity rankings: Recall that ICD_c is the main producer of NADPH. Considering sensitivities of the steady-state ICD_c flux (shown in the supplementary text: results similar to Figure 5), we find that the V_{max}'s of CIC, DIC, PC, and ME_c have the most influence over parameter interactions: their first-order rankings are low, while their total effect rankings are high. These parameters were also found to have a significant influence over parameter dependence for steady-state pyruvate concentration.

Effect of perturbation in PC V_{max} on the TCA and pyruvate cycling fluxes: As Figure 6 shows, perturbations in PC activity have their strongest effect on LDH flux, which corroborates the finding that lactate concentration increases significantly in the case of PC knock-down [8]. Considering the response in the flux through ME_c, PYC, DIC, and ICD_c to perturbations in PC activity, we see that the first-order and total-effect rankings are consistently high, indicating that the effect of PC perturbations on these fluxes is robust to changes in the values in other model parameters. These effects are compensatory: the one-at-a-time sensitivity coefficients are all negative (Figure 6d), indicating that increased flux through PC is balanced by decreased flux through these enzymes. This corroborates the findings that perturbations to PC activity do not impact the pyruvate cycling ratio and that NADPH concentrations are not sensitive to perturbations in PC activity [8].

Model Predictions: To identify target points in the model that have a strong effect in modulating pyruvate cycling ratio and NADPH concentration, we compared sensitivity rankings of the parameters across all the sensitivity methods on the pyruvate cycling ratio and NADPH concentration. The sensitivity ranking shows that the V_{max}'s of the PYC and CS have the most significant impact on both the pyruvate cycling ratio and the NADPH concentration across all the sensitivity measures. However, when we look at the rankings of the two different measures provided by the variance-based method: the total-effect and first-order rankings, we see that the rankings of V_{max}'s of PYC and CS are similar across the model outputs, indicating that these strong effects are insensitive to

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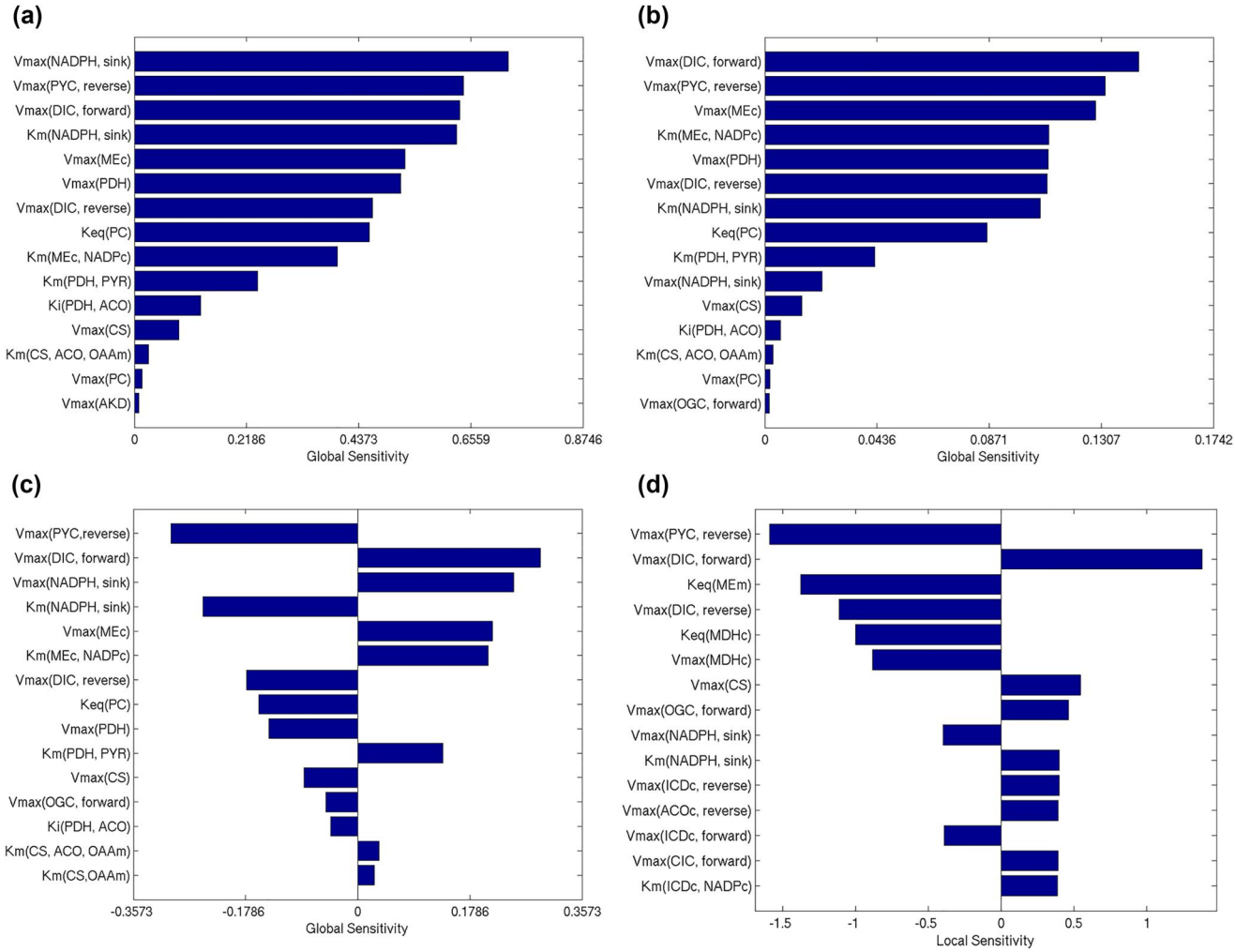

FIGURE 5 Parametric sensitivity ranking for the NADPH concentration. Panel (a) extended Fourier amplitude sensitivity test (eFAST) total effect. Panel (b) eFAST first order. Panel (c) Partial rank correlation coefficient (PRCC). Panel (d) One-at-a-time sensitivity.

perturbations in other parameters. In contrast, the V_{max}'s of transport enzymes CIC and dicarboxylate carrier (DIC), which also have significant impacts on NADPH concentration and pyruvate cycling ratio, show significant differences between first-order and total-effect rankings, and thus exert influence over the dependence on other model parameters. Knock-down of CIC and DIC has been shown to inhibit GSIS [20]. Next, we compared the sensitivity rankings of cytosolic and mitochondrial pyruvate across all the sensitivity methods. Here, we again observe that the parameters discussed above exert a strong influence on the pyruvate concentration. In addition, it has been shown that pyruvate entry into the TCA cycle through PC flux is correlated with the GSIS [1]. Therefore, we compared sensitivities of the PC flux across all the sensitivity rankings to identify target candidates for perturbing the pyruvate cycling ratio. We observe that the PC flux is influenced significantly by the candidate targets CIC and DIC in both one-at-a-time and global sensitivity ranking.

Taking these results together, the model predicts that combined perturbations (e.g., of both PYC and CIC, or PYC and DIC) can have the double-effect of a significant impact on both NADPH concentration and pyruvate cycling ratio. These combined perturbations show the greatest promise as drug targets for modulation of the pyruvate cycling ratio and NADPH:$NADP^+$ ratio and hence insulin secretion.

We simulated the model to predict the response to knock-down experiments. The model predicts that the knock-down of PYC alone has a significant impact on the pyruvate cycling ratio, but a lesser impact on NADPH concentration: when PYC activity is reduced by 10%, we see a 13% decrease in the pyruvate cycling ratio and a 5% decrease in NADPH concentration. In contrast, a 10% knock-down of DIC leads to a 12% drop in NADPH concentration and a 0.08% decrease in pyruvate cycling ratio. However, a combined knock-down of PYC and DIC by 10% each led to a significant impact on both the pyruvate cycling ratio and NADPH concentration: the pyruvate cycling ratio dropped by 30% and the NADPH concentration was reduced by 48%. This proposed combination knock-down could be addressed experimentally using siRNA knockdown, CRISPR gene-editing, or overexpression to provide further validation of the model and insight into GSIS.

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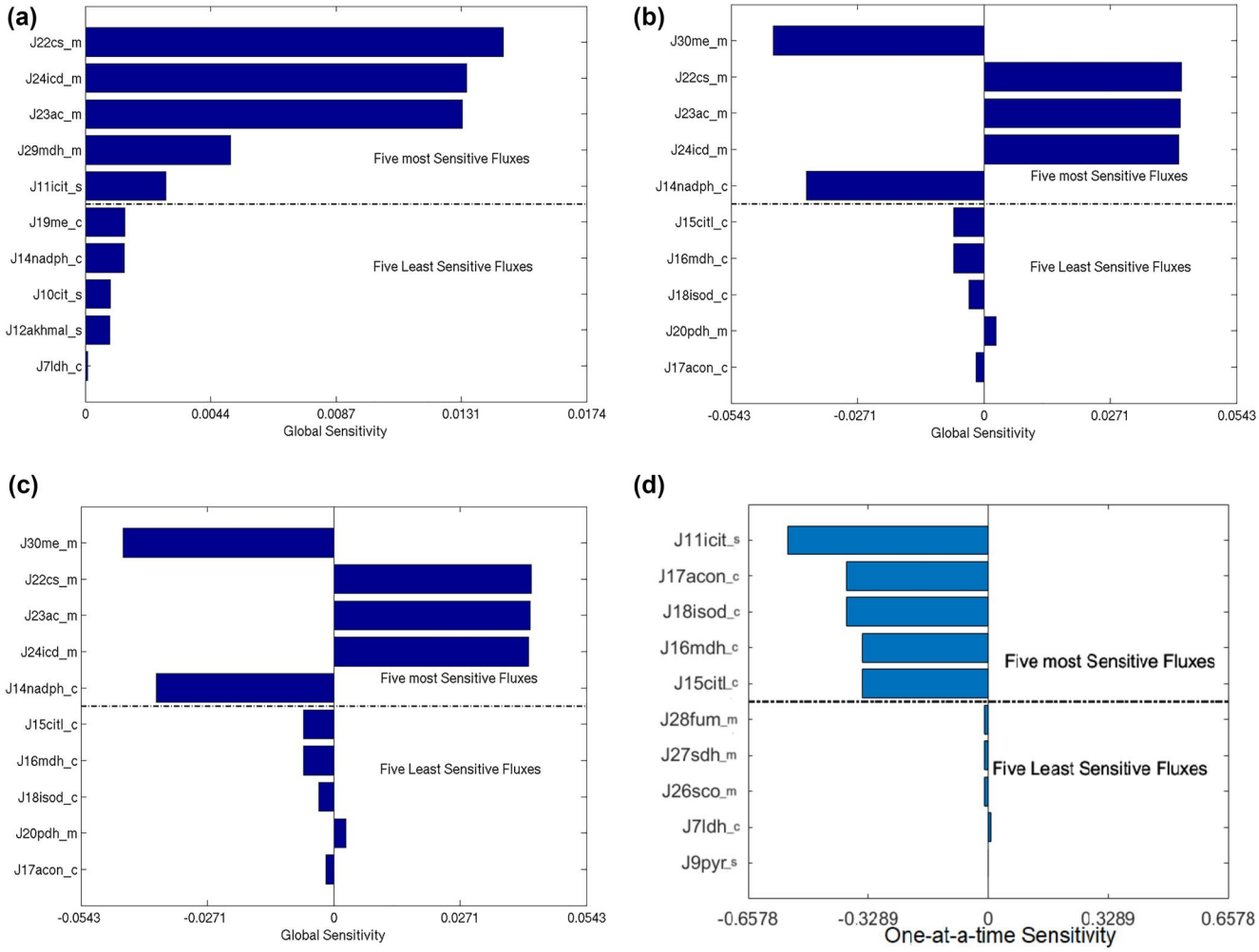

FIGURE 6 Predicted effect of perturbation in the V_{max} of pyruvate carboxylase (PC) on reaction fluxes. The five most sensitive fluxes and the five least sensitive fluxes are plotted. (a) One-at-a-time sensitivity. (b) Partial rank correlation coefficient (PRCC) ranking, (c) extended Fourier amplitude sensitivity test (eFAST) total effect and (d) eFAST first order. For flux descriptions, refer to the supplementary text.

The predicted effects can be interpreted in terms of the network structure as follows. Knockdown of PYC leads to increased pyruvate in the cytosol and reduced pyruvate in the mitochondria. The increase in cytosolic pyruvate (which is partially compensated by an increase in lactate production by LDH) results in decreased flux through ME_c and ICD_c, with a net small reduction in NADPH production. In the mitochondria, the reduced pyruvate level results in reduced PDH, PC, and ME_m flux, resulting in a modest net decrease in pyruvate cycling rate. In contrast, a decrease in DIC activity leads to increased cytosolic malate levels. Consequently, flux through ME_c increases while ICD_c flux is reduced, resulting in a reduction in NADPH levels. When combined, reductions in PYC and DIC activity elicit a nonlinear response to the jointly changing pyruvate and malate availability. NAPDH production is reduced significantly in the face of reduced cycling of pyruvate and malate into and out of mitochondria. This results from the complex interactions among ME_m (increase in flux), MDH_m, PC, and PDH (all of which decrease in flux).

The analysis presented here recommends prioritizing a number of experimental avenues. As indicated above, siRNA knock-down or overexpression experiments targeting PYC, DIC, and CIC could validate the model prediction that these are the main control points of pyruvate cycling. In addition, α-ketoglutarate is ranked highly in the sensitivity analysis. This observation recommends that manipulation of the enzymes that metabolize or produce α-ketoglutarate (ICD_c, ICD_m, and AKD) may be particularly insightful.

Moreover, experiments such as these could be used to further refine the model. The model calibration presented here is limited by that fact that data was collected from a range of cell types and using a range of assays, leading to a lack of confidence in specific model predictions, and indicating that much work is still to be done if we are to reach the goal of producing a fully predictive model of β-cell metabolism. For instance, the model presented here does a poor job of capturing the effect of ICD_c and PC knock-downs on lactate concentration. These gaps will hopefully be filled as more data becomes available, leading to model refinements that are more accurate, that will be suitable for rigorous uncertainty analysis, and that could incorporate descriptions of cell-to-cell heterogeneity or other sources of variability.

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The training and test studies described above were carried out primarily on data from rat INS-1 cell lines. These cells exhibit a broad range of GSIS sensitivity and are thus useful for investigating the amplifying pathway response. More recent studies conducted on human islets have shown similar glucose-stimulated insulin release patterns [23, 44]. In the current study, we address the pyruvate cycling pathways using the INS-1 cell line data, because the data available from the human islet are insufficient for model training. The key pyruvate cycling enzymes ME_c and ICD_c exhibit the same high activity in both rat INS-1 cells and human islets. Furthermore, as in the rat cell line, ME_m is abundant in human islet cells, underscoring the similarities of the important pyruvate cycling pathways. However, in comparison with the INS-1 cell line, human islets show reduced PC activity [44]. Previously, it has been reported that PC suppression might activate compensatory responses in β-cells either through an increasing flux of downstream pyruvate cycling fluxes such as ICD_c or through a separate pathway involving acetyl-CoA [44–46]; these effects are not explicitly included in our model. As a further extension, the model could be expanded to describe the role of α-ketoglutarate in insulin regulation and define new flux kinetics adapted to pyruvate cycling pathways [47, 48].

AUTHOR CONTRIBUTIONS

Rahul Rahul: Conceptualisation; Formal analysis; Investigation; Methodology; Software; Validation; Visualisation; Writing – original draft; Writing – review & editing. **Adam R. Stinchcombe**: Conceptualisation; Investigation; Methodology; Writing – review & editing. **Jamie W. Joseph**: Conceptualisation; Writing – review & editing. **Brian Ingalls**: Conceptualisation; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Supervision; Writing – review & editing.

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CONFLICT OF INTEREST STATEMENT

The authors have nothing to disclose.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the supplementary material of this article.

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